

SIH 2026 engineering process and readiness record

Status: PREPARATION DELIVERY - NOT SUBMISSION READY. Requested problem: SIH2171. Internal deadline: 11 September 2026, Asia/Kolkata. Prepared 8-9 September 2026 for Francis and the six-person RV University team. This records work performed and decisions taken; planned engineering is explicitly distinguished from completed implementation.

1. Scope and outcome

The requested complete entry consists of research, guardrails, measured benchmarks, a domain prototype, a submission presentation, a project website and process documentation. The authoritative identity of SIH2171 could not be established. The team therefore has a reusable preparation pack and a working static preparation website, not a verified solution to the requested statement. Domain literature, competitors, quantitative problem framing and application implementation remain blocked by that dependency.

The known team facts are the six names, institution, degree and year supplied in the brief. Proposed responsibilities are recorded in team-plan.md. No team history, photographs, contact details, registered name, registration ID or accomplishments have been inferred.

2. Intelligence collection and synthesis

Three research workstreams ran in parallel: statement identity and domain, historical competition intelligence, and design/evaluation standards. The main agent inspected the supplied PPTX, integrated the results, authored the preparation site and performed verification. Parallel scope was file-separated to avoid conflicting changes.

Source hierarchy: organiser-issued records for eligibility and requirements; original research or official technical documentation for mechanisms; sponsor/institutional announcements for awards; public participant accounts for explicitly anecdotal observations. Search snippets are discovery evidence; if a primary page cannot be retrieved, the limitation remains in the source register. A college internal-round result is not a national win. A sponsor-hosted performance claim is not independent experimental validation.

Raw preserves downloaded source text, original supplied PPTX and attributed short excerpts. Wiki contains synthesis, confidence and claim-level source links. Source registers retain original URL, author/publisher where available, retrieval time, evidence type, access limit and archived-file hash when applicable. No private WhatsApp group, Facebook group, personal inbox or restricted profile was harvested. Instagram and Facebook research produced no usable competition evidence; these are coverage gaps, not proof of absence.

Identity investigation covered official catalogue access and multiple 2024-2026 public archives, including 614 yearly statement paths in a multi-year repository. No exact SIH2171 match was found. Some official fetches were blocked; the 2025 page retrieved by the main agent had an empty statement table. Negative search evidence is bounded. A typo, internal identifier or inaccessible record remains possible, but no

possibility was selected as fact. See Wiki/problem-statement.md and Raw/problem-research.json.

The winner sample spans 2022-2025 and six institution/sponsor-documented examples. It motivates inspectable workflows, demonstrations and bounded operational claims. It cannot establish what caused winning: discovery bias, publicity bias, small sample and absent losing-team controls prevent causal or predictive conclusions. See Wiki/competition-intelligence.md.

3. Guardrails and benchmark philosophy

Guardrails are mandatory constraints with pass, fail or unknown status. Any failure rejects a release; any unknown blocks a readiness claim. An average benchmark score cannot compensate for an unverified problem, unsafe workflow or missing evidence. Preparation artifacts may still be handed over with visible labels.

The guardrail set covers scope ceilings, complete workflows, baseline-tested differentiation, prototype capacity, request validation, data authorization, deployment controls, accessibility, responsiveness, latency, Lighthouse, recovery, fallback, impact integrity, format, maintainability, attribution and real human validation. These are self-imposed engineering standards, not official SIH numerical weights.

Benchmarks define the denominator and collection method before a run. Functionality coverage is weighted acceptance-scenario completion, not code coverage. Human task completion and information findability need non-author participants. Narrative coherence needs independent ratings. API timing is separated from action-to-usable-result timing; neither a queued response nor a loading indicator counts as completion of a long task.

Judge readiness can be rated as an uncalibrated proxy. Persuasion probability remains null because no comparable outcome-labelled dataset and held-out calibration exist. A readiness score of 85 is not an 85% chance of winning. Domain social-impact values also remain null until the beneficiary, baseline and observation unit are defined.

Saturation requires all mandatory gates to pass, objective targets to be met, human evidence to exist, and two evaluated iterations with marginal improvement below the declared thresholds. The deadline does not turn unknown into pass. Saturation has not been achieved.

4. Technology selection and feasibility

The conditional application architecture is one web interface, one API/domain service and at most one primary database. React/TypeScript plus Node can support a conventional web workflow with shared types and a simple development path. SQLite suits a single local demo instance; Postgres is more appropriate for a shared multi-instance deployment. Python/FastAPI may supersede this choice if scientific or ML libraries define the problem. Hardware requirements would invalidate a web-only plan.

No stack is final until statement verification. No domain application, server API or persistence has been implemented. Prototype/README.md records the build contract, complete-flow requirements, error cases, deployment and fallback acceptance. Creating a generic dashboard would create code without proving problem fit.

The actual website uses semantic HTML, local CSS and small JavaScript with optional locally served GSAP motion. There is no build dependency, external font request, database, analytics or personal-data form. It can run offline after local delivery. Decisions and alternatives are in architecture-decisions.md.

5. Design decision audit

The requested Taste skill and primary Awesome DESIGN.md collection informed a written design baseline. The original visual direction uses paper, ink and signal lime, large typography, chapter spacing, ruled evidence lists, an original abstract path illustration and typographic team monograms. Lando Norris's site was a reference for hierarchy and personality; no brand assets, photographs, signature or proprietary font were copied.

The site's first screen explicitly says the ID is pending and the domain prototype is not built. This prevents the presentation style from implying implementation. The navigation leads to evidence, proposed architecture, team ownership, delivery milestones and downloads. Team initials remain honest placeholders for authentic portraits. A share action explains why a local preview is not publicly shareable. A contact channel is labelled pending; no message-success fiction is used.

Taste suggestions for stock imagery, testimonials, always-on motion and rigid random components were subordinated to relevance, truthfulness, accessibility and performance. Real seeded selection guided layout exploration. Feedback carousel became a user-operated milestone carousel; no invented feedback was added. Animated content stays visible without JavaScript, and reduced motion disables the optional motion path.

6. Iteration and correction log

Iteration 0 established empty workspace structure and source provenance. Template XML revealed six content slides and a seventh instruction slide. The requested 12-18-slide format was resolved into two separately named artifacts: a six-slide submission blueprint derived from the supplied file, and a 15-slide expanded pitch blueprint. Current-year validity is unverified because the supplied file is dated 2025.

Iteration 1 produced the static website with evidence disclosures, interactive proposed architecture and manual milestone controls. Desktop inspection at 1440 pixels showed no page-level horizontal overflow. Review identified that readiness status initially sat below the hero; explicit pending status was added inside the first-screen copy. Monograms received image roles and descriptive accessible names.

Final local Lighthouse medians were 100 in every category on both profiles, across six runs. A visible-label mismatch was corrected and retested. Document exports and further browser checks are logged separately in verification.md. Any later correction must name the failed observation, changed file and retest. There is no fabricated domain prototype iteration history: implementation did not begin without a statement. Automated performance and browser results apply only to the preparation website.

7. Accessibility review approach

WCAG 2.1 AA is a target, not a certification. The preparation site uses semantic landmarks, a skip link, one primary heading, native expandable evidence sections, real buttons, visible focus, text-based statuses, descriptive image alternatives, and reduced-motion support. Architecture selection uses pressed states and a polite live region. Milestones do not auto-advance.

Automated audits find only part of the criteria. Full conformance still needs criterion-level review, keyboard and screen-reader checks, 200% zoom/text-spacing review, contrast in all states and complete-process testing. No full conformance claim is made. Reference: <https://www.w3.org/TR/WCAG21/> . Completed checks and remaining limitations are in verification.md.

8. Performance and reliability

The static site avoids remote runtime requests and large media. SVG art is original and small; scripts are local, deferred and optional for reading the content. Lighthouse measurements must retain every report and exact device profile. The declared gate uses three runs per profile, median strictly above 90 in all four categories. This is a website audit, not proof of a domain application meeting the under-200ms core-flow requirement.

Domain performance requires at least 100 measured attempts after warmup, recorded environment and per-flow p95. Capacity requires the fixed-session/data-size test described in Guardrails. These domain tests have not run. For the eventual demo, freeze seed data, verify restart persistence, rehearse three complete runs and record the exact tested build. A local video and captions must be played without internet before being called a fallback.

9. Security by design

The delivered static website has no authenticated state or data collection. It does not contain API keys or an operational contact endpoint. Downloads are intentionally public preparation artifacts. The development server binds to loopback. Deployment instructions require HTTPS and suitable response headers, but public deployment has not been verified.

The future API must validate all incoming fields, bound payloads and pagination, parameterise queries, control output encoding, authorize private records server-side and test cross-user access. Duplicate submissions need idempotent behavior when applicable. Store credentials only in local or host configuration; redact sensitive logs. Threat modelling must be tailored to the verified domain. This design contract is not security-testing evidence for an app that does not exist.

10. Sustainability and social impact

The engineering sustainability strategy favors few services, reproducible setup, local demo capability, explicit ownership and a documented recovery path. Small assets reduce transfer and dependency exposure, but no energy or carbon savings have been measured. Avoid custom model training unless a baseline comparison proves it necessary.

Impact-model.md specifies a baseline, comparable task, unit, beneficiary, observation period, counterfactual and uncertainty approach. Operating cost must include hosting, storage, third-party usage and support effort under labelled workload assumptions. No purchaser, funding route, partnership, revenue estimate or adoption rate is claimed. The business model depends on the actual domain.

11. Gap resolution and final self-evaluation

Release is blocked by unresolved statement identity and missing domain implementation. Additional gaps are authentic team assets, registered details, contact destination, current-year submission instructions, measured domain performance, human user/reviewer studies, pilot validation and a real demo recording. These are not permission requests; they are facts about available evidence.

The historical judging dimensions are only partially addressed. Clarity has a source ledger and blueprint; feasibility has a conditional architecture; UX has a functioning preparation website. Novelty, practicability, domain impact, sustainability validation and demonstrability remain unproved. No aggregate competition

score or selection probability is assigned.

To unblock: preserve an organiser-issued full problem statement with exact ID/year, sponsor, category, expected solution and constraints. Reassess the stack, freeze the primary journey, compare existing approaches, implement and test the vertical slice, then replace all blueprint evidence slots. Confirm registration and actual submission format before upload. This preparation pack reduces setup work; it does not substitute for those steps.

12. Working reference

- Start with Wiki/problem-statement.md for identity evidence and limitations.
- Use Raw source registers to verify attribution; Wiki provides synthesis.
- Read DESIGN.md before editing the interface.
- Keep Guardrails authoritative; store run evidence under Benchmarks/results.
- Run scripts/check_release.py to compute readiness; a nonzero exit is expected while blocked.
- Run the website using the command in Website/README.md.
- Edit the supplied-format submission blueprint separately from the expanded talk.
- Present targets as targets, projections as projections and measured results with their test conditions.